

Enterprise integration using the agent paradigm: foundations of multi-agent-based integrative business information systems

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Received 5 December 2003; received in revised form 23 September 2004; accepted 26 September 2004

Available online 27 October 2004

Abstract

Enterprise integration through integrated business information systems (IBIS) is necessary to achieve agility in the current age of hyper-competition. Multi-agent systems (MAS) provide a new paradigm for IBIS development. In this paper, we review the IBIS modeling and MAS literatures and find that the MAS paradigm provides an excellent approach for modeling and implementing IBIS systems. We synthesize these two bodies of literature and propose a conceptual framework for multi-agent-based integrative business information systems (MIBIS) and a unified set of eight orthogonal ontological constructs that are minimally required for any conceptual modeling grammar for the MIBIS bounded universe of discourse.

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Keywords: Multi-agent systems; Enterprise integration; Coordination; Integrative business information systems; Multi-agent-based integrative business information systems; Systems analysis; Systems modeling; Conceptual modeling

1. Introduction

To thrive in the current hypercompetitive environment, businesses not only need to integrate their internal stovepipe applications, they also need to

integrate their application systems with their supply chain partners' systems. Both the practitioner publications and academic literature have noted the significant benefits that information systems integration both within and across the enterprise can bring about for businesses in terms of improved planning, timely deliveries, reduced inventories, reduced costs, improved product line in tune with market needs, and responsive and improved customer service.

Information systems (IS) reengineering and integration became one of the most important IS issues in the early 1990s, driven by the call for business process

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simplification and cross-functional process integration, the key tenets in Hammer and Champy's [54] landmark work on business process reengineering (BPR). In fact, the top four IS issues that emerged from a 1994–1995 Delphi study of senior IT executives [12], pertain directly to the notions of IS responsiveness, reengineering, and integration.² Work in these inter-related areas has continued since the early 1990s in both the information technology (IT) industry and the academia under a variety of labels including enterprise resource planning (ERP), enterprise application integration (EAI), integrated supply chains, and workflow management.

Most of the enterprise and IS integration efforts have utilized the object-oriented (OO) paradigm and component-based architectures as the technological solution for the integration problem (e.g., Refs. [4,34–36,67,81,115–117]). However, some researchers have focused on the intelligent agent and multi-agent systems (MAS) approaches as more suitable alternatives for e-business and enterprise integration applications, and have developed and utilized agent approaches and technologies in e-business applications [66,70,72,124,142], business process management [63,65,102], supply chain management [60,118,143], enterprise integration [82,101,110,111], and manufacturing [76,109].

While the popularity and application of the agent technology in the business domain has grown over the recent years, the field is currently marked by unique and innovative approaches and architectures for solving the business and IS integration problem. There is currently a lack of a unifying framework that not only synthesizes literatures in the two pertinent streams—business/IS integration (e.g., BPR, ERP, workflow, etc.) and the MAS paradigm—but also provides a foundation for conceptual analysis and modeling of integrative business information systems based on the multi-agent systems paradigm. The goal of the present paper is to fill this void. We review relevant literature in these two key major areas and find that the MAS paradigm indeed provides an excellent approach and suitable mecha-

nisms for developing integrative business information systems to achieve the goal of creating an integrated enterprise. We also synthesize the two bodies of literature and propose a conceptual framework for multi-agent-based integrative business information systems (MIBIS).³ We also identify a minimal set of orthogonal ontological constructs [15,73,114] that are central to the MIBIS bounded discourse universe [74].⁴ We take the approach of minimal ontological commitment [51] in our synthesis as we wish to identify only those ontological constructs that are absolutely essential for conceptual analysis and modeling of MIBIS systems.

The paper is organized as follows. In Section 2, we discuss the role of information systems in business integration, define the concept of Integrative Business Information Systems (IBIS), and review a number of approaches authors have taken for conceptual analysis and modeling of such systems. In Section 3, we discuss the notion of agents, agent communication, multi-agent systems, and how multi-agent systems provide an appropriate architecture for IBIS using various coordination mechanisms. In Section 4, we synthesize the IBIS and MAS literatures, and develop a unifying multi-agent-based integrative business information systems (MIBIS) framework as a means for achieving business integration. In Section 5, we discuss why the constructs are necessary for the analysis and design of MIBIS systems and how they will help enterprise integration modeling. Finally, Section 6 concludes the paper.

³ While the acronym MIBIS contains the letter S for the term system, we use the phrase MIBIS system for ease of reading. We also use the phrase MIBIS entity in this paper to refer to a multi-agent-based integrative business information system.

⁴ A universe of discourse comprises the realm of the phenomenon of interest and of the logically possible propositions within that realm. A bounded discourse universe delineates a narrower realm of interest. If information systems is treated as an unbounded universe (because it will deal with concepts from every imaginable perspective), the MIBIS universe is a bounded universe because we are only interested in concepts pertaining to a particular type of information system, one which is an integrative business information system and is based on the multi-agent systems paradigm.

² The top four issues in this study include: (1) building a responsive IT infrastructure, (2) facilitating and managing business process redesign, (3) developing and managing distributed systems, and (4) developing and implementing an information architecture.

2. Integrative business information systems

In this paper, we follow the approach taken by Alter which suggests that work system is the highest-level concept within organizations, and information systems are only a particular type of work system that support other organizational work systems. Adopting Alter's [3] definition, we define a work system as "a system in which human participants and/or machines perform a business process using information, technology, and other resources to produce products and/or services for internal or external customers." A business organization typically contains a number of work systems (such as for procuring materials, recruiting and retaining employees, sourcing technologies and other resources, selling products and services, etc.) and uses them to conduct its day-to-day business operations. Business integration essentially involves the integration of multiple work systems within a business organization.

An information system, as a particular type of work system, supports or serves other work systems within organizations [3]. Fig. 1, taken from Ref. [2], depicts the overlap between work systems and information systems. This figure shows how information systems can support business integration by integrating multiple work systems through their support of business processes in more than one work system (as in scenarios D and E in Fig. 1). We refer to these types of information systems as Integrative Business Information System (IBIS).

2.1. An overview of IBIS and IBIS-supported enterprise integration

An IBIS is generally a complex information system that integrates multiple work systems in a single information system. As discussed earlier, a variety of IBIS types have been developed in the 1990s under various labels. For example, ERP is an enterprise-wide IBIS that integrates cross-functional processes and streamlines information flow across different enterprise functions in an organization such as product planning, purchasing, inventory management, customer service, etc. [79]. EAI, another approach to IBIS, involves the integration of various enterprise applications so that they can share information and processes freely [85]. An integrated supply chain is also supported by an IBIS

which tightly connects supply chain partners of an enterprise spanning multiple enterprise functions from the ordering and receipt of raw materials through the manufacturing, distribution, and delivery of products to customers. Workflow systems, another type of IBIS, enable "the automation of a business process, in whole or part, during which documents, information or tasks are passed from one participant to another for action, according to a set of procedural rules" [38].

An IBIS is different from a traditional information system in that the business problems addressed by an IBIS are quite complex. Generally, a single work system or a single actor within a work system does not have the necessary knowledge, resources, and capability to solve such problems. Further, while work systems in organizations are generally connected, they are relatively autonomous, have their individual goals, perform their own tasks to achieve their goals, and control how their resources are consumed, by whom, at what cost, and in what time frame. The role of IBISs is to integrate different work systems by helping solve problems that span multiple work systems and by coordinating the goals, tasks, and resources of multiple work systems in order to achieve the overall system goals.

The above view of IBIS is illustrated in Fig. 2. A business enterprise consists of a number of work systems each of which has been assigned certain goals to achieve that are derived from enterprise goals. It should be noted that the business enterprise shown in the figure can be a single business organization or it can be an extended, networked, or virtual organization or a strategic alliance comprising of a number of interacting partner firms who wish to integrate some of their work systems. Each work system contains multiple actors who are assigned sub-goals derived from work system goals, and who perform a number of tasks using some resources to achieve their goals, and in the process use or create some information objects. Accomplishment of actor goals contributes to the accomplishment of work system goals, which in turn contributes to the accomplishment of enterprise goals.

Enterprise integration requires the integration of work systems within a business organization. The links shown between individual work systems in Fig. 2 capture and portray important requirements for integrating multiple work systems within or across organizations. An IBIS needs to support the integration

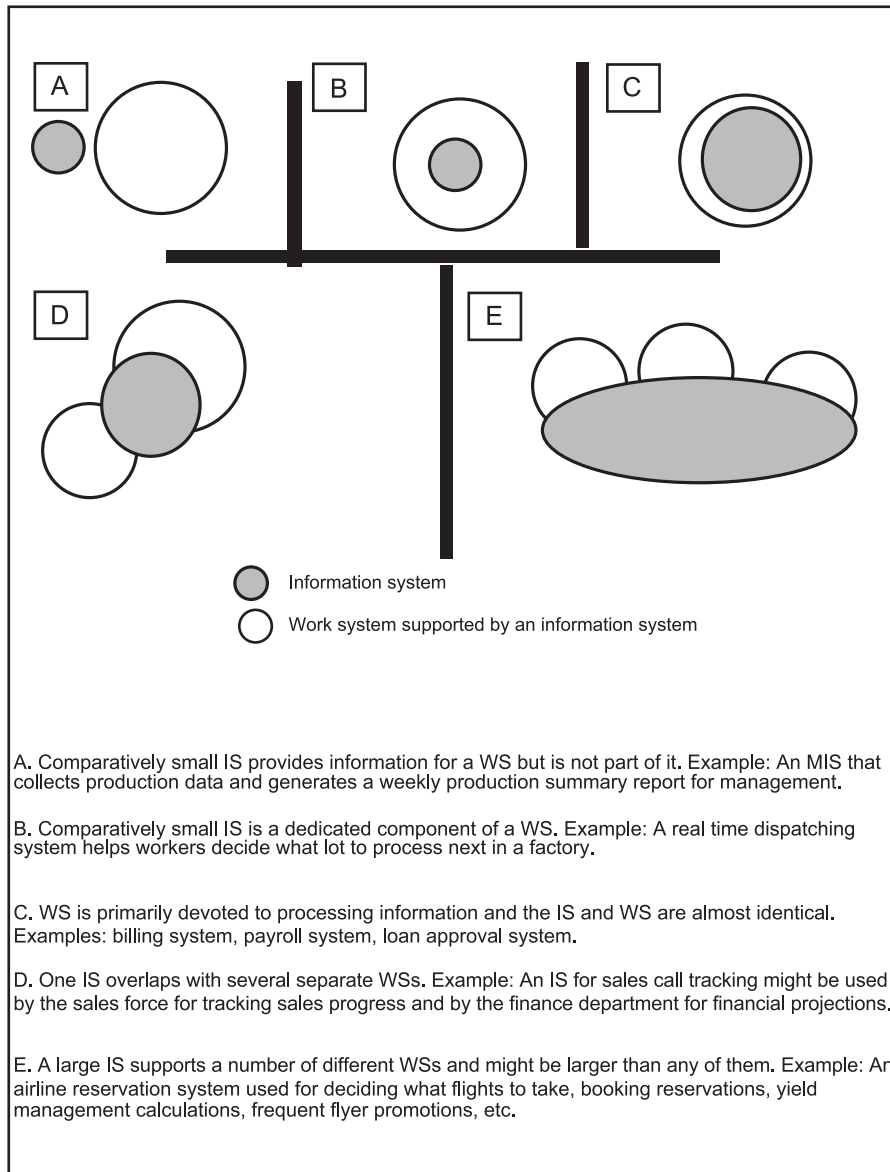


Fig. 1. Various types of overlap between information systems and related work systems (adopted from Ref. [2]). Note: Scenarios D and E depict Integrative Business Information Systems.

requirements shown in Fig. 2—work systems integration, data integration, and technology integration—in addition to providing task and decision support within individual work systems. However, this paper only addresses work systems integration, as it is this integration that is concerned with the key organizational aspects of goals, tasks (business processes), actors, resources, etc. While data and technology

integration are foundational requirements for realizing the IBIS vision, for the purposes of this paper, we assume that data and technology integration are supported by the modern-day Internet, web, and database technologies, and the open standards that enable integration across a variety of technological platforms. Data and technology integration are, therefore, not covered in this paper.

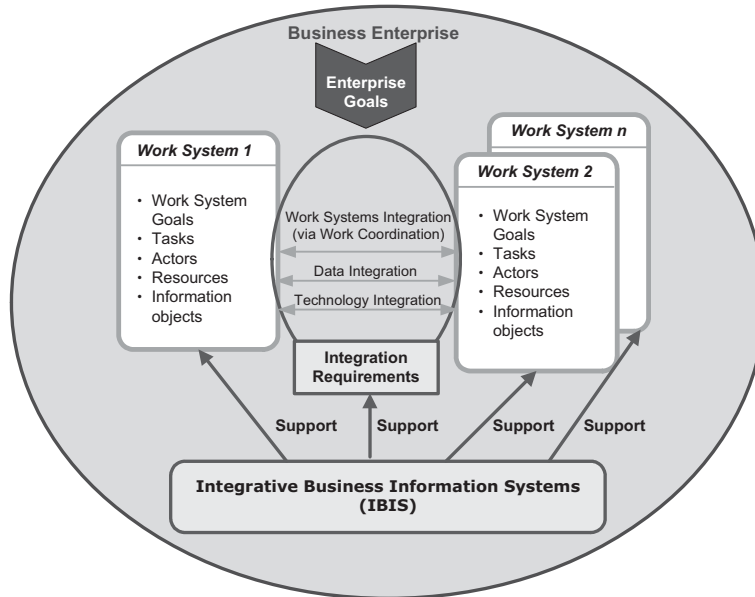


Fig. 2. An overview of work systems and integrative business information systems (IBIS). Note: The business enterprise shown in the figure can be a single business organization or it can be an extended/networked/virtual strategic alliance comprising of a number of interacting partner firms who wish to integrate some of their work systems.

Work systems integration involves structuring workflow, allocating tasks and resources to actors, sharing resources, managing interdependencies (or simply dependencies) between tasks, resources, and actors, and enabling communication across multiple work systems. These are all issues that have been discussed as part of the notion of coordination in the literature [21,87]. In order to design and implement IBISs that can effectively support work systems integration, these coordination issues need to be properly analyzed and modeled. We, therefore, first review the fundamental concepts of coordination below and then review some of the major IBIS analysis and modeling techniques highlighting the particular integration areas they address and the approach they take. We divide our review of IBIS analysis and modeling techniques under two major categories—dependency/workflow perspective and communication/conversation perspective—based on how the literature has developed in this area. It should be noted that these two perspectives to work systems coordination include substantial overlaps and we have classified research under a particular perspective based on our assessment of the focus of specific modeling techniques.

2.2. IBIS analysis and modeling

2.2.1. Work systems coordination

Coordination is at the heart of business organizations and work systems as it is one of the fundamental principles for organizing work and managing complexity in social systems and business organizations. Some of the earliest notions about coordination were developed by systems theorists (e.g., Refs. [5,71]), which were later refined by organizational theorists notably (e.g., Refs. [14,42,78,125]). Thompson's [125] three organizational coordination mechanisms—planning, standardization, and mutual adjustment—proposed over three decades ago are still well-accepted and used in organizational research (e.g., Ref. [52]).

More recently, Malone and Crowston [87] have developed an interdisciplinary coordination theory drawing from various disciplines including computer science, organization theory, operations research, economics, linguistics, and psychology. They define coordination as “managing dependencies between activities” based on the notion that if there are no interdependencies among activities, there is no need for coordination in the system [87]. These authors also identify several common dependencies and analyze

Table 1
Dependencies and coordination mechanisms from Ref. [87]

Dependency	Examples of coordination mechanisms
Shared resources	“First come/first serve”, priority order, budgets, managerial decision, market-like bidding (same as for “Shared resources”)
Task assignments	
Producer/consumer relationships	
Prerequisite constraints	Notification, sequencing, tracking
Transfer	Inventory management
Usability	Standardization, consulting users, participatory design
Design for manufacturability	Concurrent engineering
Simultaneity constraints	Scheduling, synchronization
Task/subtask	Goal selection, task decomposition

coordination mechanisms to manage them. We reproduce their dependencies and coordination mechanisms framework in Table 1 to provide a basic understanding of the concepts of coordination.

In related work, Crowston [20,21] provides a typology of dependencies and coordination mechanisms based on the objects involved in the dependency, and it includes task–task, task–resource, and resource–resource dependencies and associated coordination mechanisms. Coordination subsumes the notions of workflow because they essentially deal with issues of task–task and task–resource dependencies and their coordination. Further, Malone and Crowston [87] also identify the processes that underlie coordination, and suggest that coordination is the highest-level process that requires some kind of group decision-making. This in turn needs communication that involves the notions of senders, receivers, messages, and languages.

Similar kinds of notions about coordination have also been proposed by Holt [55]. According to him: “...Coordination has no product. Instead it serves to establish relationships between tasks and their products. Coordination has no independent purpose; it is a prerequisite for the accomplishment of other purposes.” He also provides examples of some types of coordination actions necessary to be undertaken, including:

- *Safekeeping* of parts when they are required to be used in some other tasks;

- *Synchronization and Transfer* to insure that two interdependent tasks can be undertaken without much loss of time; and
- *Acceptance* of the person receiving the parts to the effect that they indeed meet his/her requirements.

Based on the above discussion, we treat coordination as the glue that binds work systems together. For effective work systems integration, therefore, coordination requirements need to be fully supported by integrative business information systems. In the following sub-sections we review some relevant analysis and modeling techniques that focus on and support one of the three coordination perspectives: managing dependencies, managing workflow, and managing communication among various work systems.

2.2.2. The dependency/workflow perspective

The dependency perspective presumes that coordination problems are caused in the first place by interdependencies among various organizational actors, tasks, and resources, and that an effective coordination system should address and resolve these mutual interdependencies.

The Actor Dependency Model proposed by Yu and Mylopoulos [140] is an attempt to capture the intentional dimension of organizational work. The authors propose that actors within organizations are autonomous with their own desires and abilities, and they use their know-how and resources available to them to pursue their goals. They further propose that business processes are accomplished through a network of interdependencies among actors who depend on each other for goals to be achieved, tasks to be performed, and resources to be produced and utilized. Actors depend on each other for accomplishment of certain goals (goal dependency), for completion of certain tasks in specific ways decided by actors who depends on other actors (task dependency), for availability of some common physical or informational resources (resource dependency), and for accomplishment of some goals that are not sharply defined but need clarification between two actors (soft-goal dependency). These authors further use an “Issue Argumentation” model for reasoning about process redesign that is based on the concepts of non-functional requirements analysis in software engineering [16,95]. The Actor Dependency model provides a

graphical approach for modeling which is both intuitive and simple. This approach draws upon the concepts of dependency from organization theory [104,125]. This model is helpful in defining the flow of work based on dependencies of various elements that are inherent in organized work within organizations.

Recently, Tillquist et al. [126] has developed dependency network diagrams (DNDs) to represent the essential elements of inter-organizational relationships based on resource dependency theory [105]. A DND diagrammatically represents an organizational information system as a network of dependencies among a set of roles. Role is defined in this approach as a combination of specific set of activities and goals, which the activities will accomplish. A dependency of one role on another role is then defined as an interface relation between the two roles in terms of activities that need to be performed by the latter role for achievement of a subset of goals of the former role. By analyzing and formally depicting these dependencies, DNDs can guide the design of information systems to explicitly control and coordinate organizational activities. The DND syntax and method is quite straightforward and can be used not only to diagnose an existing system but also to prescribe an ideal future system.

The “Easily CHangeable Office Systems (ECHOES)” visual modeling language [92] addresses some important aspects of integrative business information systems. The ECHOES technique integrates the information flow aspect, the service description aspect, the information description aspect, and the organizational aspect (specifically an actor’s job description and services that can be performed by the actor). The model is based on a multi-paradigm visual language and provides the end user, who is presumed to be a domain specialist, with direct visual languages for modeling the different facets of an office. It is worth mentioning here that even though the modeling scheme provides different modeling languages for the different facets of the office, the models are semantically integrated through a common underlying model of the office embedded within the ECHOES paradigm.

Role Activity Diagrams (RADs) are a set of graphic notations for modeling business processes. The basic concepts of RAD were first introduced by Holt et al. [56], and later enriched by Ould [99]. The

RAD technique partitions a business process into a set of interacting roles. Each of these roles group activities together that might be carried out by a person, group, or machine (an actor or an agent). Roles have constructs to depict concurrent or parallel behaviors. They act in parallel, and communicate and synchronize through interactions. RAD is essentially a state-based modeling technique and actions and interactions of a role move it from the current state to a new state.

The Workflow Intelligent Business Object (WIBO) approach is devised to model and implement a workflow with intelligence, autonomy, collaboration and cooperation [33]. The approach models a workflow as a collection of interacting business objects that are able to manage themselves. These objects belong to four meta-level classes or types—process, role, actor, and resources. When a workflow is modeled using the WIBO approach, activities that form a process are first allocated to roles; the roles then assign those activities to actors. The actors schedule the activities by allocating resources. If an activity is not performed before its deadline, the actor responsible for the activity is alerted; if the activity is still not executed after this alert, the process considers its reallocation to a different actor. In this manner, the WIBO approach promises a better dynamic management of processes and utilization of resources.

eXchangeable Routing Language (XRL), based on XML, is a recent language developed to support seamless routing of inter-organizational workflow [127]. XRL is designed to suit both distributed and asynchronous modes of operation, and it supports the transfer of not only transaction data but also information describing workflow processes at an instance level. The semantics of XRL are expressible in terms of Petri Nets, and all XRL routing elements can be automatically translated into Petri net constructs. A routing element is a building block to describe a task, task sequence, parallel condition, condition checking and looping, an event, or a task state. An XRL schema is a collection of routing elements. The authors also propose an architecture for developing workflow management systems based on XRL.

Bajaj and Ram [8] suggest that most workflow modeling techniques lack capabilities to capture different aspects of a workflow system such as data, process, and organization in a single consistent view. They thus

propose State–Entity–Activity–Model (SEAM) using set theory to overcome this shortage. The model explicitly links activities and data via states. The authors also introduce temporal concepts into the model. SEAM is applied to a real-life organization to show how the SEAM schema can be implemented on a relational database management system.

Basu and Blanning [9] also propose a framework to integrate the informational, functional, and organizational perspectives of the workflow in a single model. The workflow modeling is based on a powerful graph-theoretic construct, a metagraph, which is a graphical structure that represents directed relationships between sets of elements. A workflow can be represented as a metapath from a set of information elements comprising a source to another set comprising the target. It should be noted that metagraph-based workflow modeling provides not only graphical visualization of processes, but also their formal analysis to formulate questions about the relationships among information, tasks, and resources.

Fox et al. [41] propose an organizational ontology for enterprise modeling. The authors consider an organization to be a set of constraints on the activities performed by organizational agents. An agent plays one or more roles. Each role is defined with a set of goals that the role is created to fulfill and is allocated with proper authority at the level that the role can achieve its goals. Agents perform activities in the organization, each of which may consume resources and there is a set of constraints that constrain agent activities. An agent can also be a member of a team set up in response to a special task, possesses skills, and has a set of communication links that specify the other agents in the organization with who it can communicate.

2.2.3. *The communication and conversation perspective*

Communication and conversation among actors are obviously very important coordination mechanisms in any business process/work system. They become all the more important in the context of cross-functional/multiple work systems integration. In the context of organizational work systems and IBIS, communications and conversations have been analyzed and modeled since the 1980s using a new paradigm called the language/action perspective (LAP). The LAP has

emerged from the pioneering work of Flores and Winograd [39,40,135,136] and is grounded in the Speech Acts Theory [7,107,108] and the Theory of Communicative Action [53]. This perspective proposes that an action is performed when someone speaks or converses with someone else in an organization. The individual speech acts such as requests, commitments, promises, acceptance, cancellation, declaration, etc., give rise to larger conversation structures that may affect one's future courses of action. This research suggests that communications and conversations conducted using language are the basic drivers for action within human organizations. The language/action approach, therefore, views conversations for action as the key structure that need to be analyzed and modeled for successful design of information systems.

One of the first applications of the language/action approach in workflow modeling is an Action Workflow System [91]. This approach models every action in a workflow as a four-phase loop based on communication between a customer and a performer, including a proposal (which may include requests by a customer or offers by a performer), an agreement on the conditions of satisfaction, the performance of the action itself, and a declaration of satisfaction by the customer. While the Action Workflow approach is highly intuitive and simple, some authors suggest that it is not appropriate for modeling business processes that have objectives other than customer satisfaction [46]. The approach also suffers from limitations of complexity management as it does not provide appropriate means for developing interdependent and hierarchical workflow loops.

The SAMPO modeling technique [6] focuses on the discourses that go on in offices, and considers discourse as a sustained stretch of speech. This modeling technique proposes the notions of entities, acts, and agents, agents being entities that can perform various types of acts. The acts themselves are divided into instrumental acts and speech acts; the instrumental acts are physical human deeds and speech acts are symbolic deeds that result in linguistic expression having a meaning. The authors suggest that out of a number of other types of speech acts it is the illocutionary speech acts that are meaningful for analysis of office information systems.

Dynamic Essential Modeling of Organizations (DEMO) [27,28,129] is a formal and rigorous organizational and business process modeling technique rooted in the LAP. This approach introduces two types of conversations: informative conversations only reproduce already existing information, whereas performative conversations change the state and create a new agenda for action. Performative conversations are further divided into factagenic (those that generates some new facts) and actagenic (those that generate some action) conversations. A business transaction is then defined as the combination of an actagenic conversation, an essential action, and a factagenic conversation. DEMO includes five partial models for conceptualizing an organization as a network of business relation transactions [128]. These models are the interaction model, the business process model, the action model, the fact model, and the interstriction model. The interaction model specifies the types of business transactions and the actors involved. The business process model specifies the order and the conditions under which the business transactions are executed. The action model includes the detailed specifications of the operation of a business system. The fact model provides complete and precise specifications of all the information that is created and/or used by the system. Finally, the interstriction model provides the identification and the specification of the informative conversations between actors and information banks in the system.

Weigand et al. [133] propose a layered pattern of definition language for electronic commerce transactions modeling, based on the various LAP approaches. Their framework includes five layers. In the first layer, speech acts are regarded as atomic building blocks of business processes. In the second layer, a transaction is defined as the smallest possible sequence of actions (speech acts) that leads to a certain deontic state. In the third level, a workflow is defined as a set of related transactions aimed at achieving some goal. In the fourth layer, the notion of contract is introduced to capture a symmetrical type of exchange, in which all actors involved in the conversation have a common interest in a particular object. And in the last layer, a scenario is described as the interactions that take place among several concurrently running contracts among several parties.

Table 2

Summary of key constructs in IBIS analysis and modeling techniques

IBIS modeling technique	Key constructs
Actor Dependency Model	Actors, Know-how, Resources, Business Processes, Goals, Tasks, Interdependencies,
Dependency Network Diagrams	Dependencies, Roles, Goals, Activities
ECHOES Modeling Language	Information flows, Services, Information, Actors
Role Activity Diagrams	Business Processes (Concurrent versus Parallel), Roles, Activities, Actors/Agents, Interactions
Workflow Intelligent Business Object	Business Objects, Interactions, Processes, Roles, Actors, Resources, Activities, Workflow
XRL	Workflow processes, Routing Elements, Tasks, Task Patterns, Events, States
Action Workflow System	Action, Actors (Customer and Performer), Four-Phase Workflow Loop (Proposal, Conditions of Satisfaction, Performance, Declaration of Satisfaction)
SAMPO Model	Entities, Acts (Instrumental versus Speech Acts), Agents
DEMO Model	Conversations (Informative versus Performative Conversations; Factagenic versus Actagenic Conversations), Actions, Actors, Business Transactions (combination of an actagenic conversation, an essential action, and a factagenic conversation), Information Banks
Weigand's Model [133]	Actors, Actions (based on Speech Acts), Business Processes, Transactions (Sequence of Actions), Workflow (Set of related Transactions), Goals, Contracts (Symmetrical type of Transactions), Conversations (among actors), Interactions (among several concurrently running contracts)
Lind and Goldkuhl's Model [84]	Business Acts (Speech Acts versus Material Acts), Business Interactions, Business Action (Pair of Business Acts), Business Exchanges (Collections of Actions), Business Transactions (Collection of Business Exchanges)
SEAM TOVE	Entity, State, Activity, Workflow Organization, Goals, Division, Roles, Agents, Teams, Activities, Resources, Communication links, Skills, Authorities, Constraints
Basu and Blanning's Model [9]	Information, Task, Resource, Workflow

Another framework for business modeling based on the LAP is proposed by Lind and Goldkuhl [83] to supplement the framework of Weigand et al. This framework also comprises five layers. These authors suggest using business acts (a collection of speech acts and material acts) as the basic unit of analysis in the first layer because a business interaction includes not only speech acts but also material acts. In the second layer, they use a more general concept—an action pair—rather than a transaction. An action pair is defined as the occurrence of one business act functioning as a trigger for another act, which will have the function of a response. In the third layer, the notion of exchange is introduced, which means that actions in business are always reciprocal. The fourth layer, business transaction, is built from different types of exchanges related to one another. In the fifth layer, related business transactions between two actors are formulated into a transaction group.

In summary, the above review indicates that IBIS analysis and modeling techniques utilize a wide variety of constructs to model integrative work systems and these are summarized in Table 2. While each technique utilizes a unique approach to model what it believes are the essential elements of integrative work systems, several concepts across these techniques are also common. These common, and arguably essential, constructs include goals, roles, actors (or agents), activities (or tasks, processes), interactions (or conversations), workflow, resources, information, and of course interdependencies. We now turn our attention to the multi-agent systems paradigm in the following section.

3. Multi-agent systems

Multi-agent systems are gradually becoming a new paradigm for developing distributed computing systems. This paradigm provides an appropriate architecture for the design and implementation of integrative business information systems as it addresses adequately the crucial requirements of coordination which, as discussed above, is central to the IBIS paradigm. In this section, we provide an overview of software agents, discuss multi-agent systems as coordination models, and then discuss

various coordination mechanisms in multi-agent systems.

3.1. Agents and agent communication

While there is no universally agreed definition of a software agent,⁵ the following definition is most widely accepted: “an agent is a computer system that is situated in some environment, and that is capable of autonomous action in this environment in order to meet its design objectives” [134,138]. Further, it has been proposed that an intelligent agent is autonomous, reactive, proactive, and social [138]. An agent is argued to be different from a traditional object. First, agents are commonly modeled using “mentalistic” notions, such as knowledge, belief, intention, and obligation [106], while objects are modeled as simply encapsulating their internal structure as methods and attributes. Second, the degree to which agents and objects are autonomous is quite different. Objects do not have control over their behaviors, because they are invoked by others. On the contrary, agents are able to decide whether or not to execute an action after receiving requests [137]. In addition, unlike objects that just engage in single-message exchanges, agents engage in conversations that are task-oriented shared sequences of messages [77].

The primary difference between agent and object communications lies in the language of the interface [44]. In general object-oriented programming, the meaning of a message can vary from one object to another. On the contrary, in an agent-based system, agents use a common language with agent-independent semantics. There are different types of messages an agent can receive, and the agent is able to evaluate the received messages and decide what actions to perform. Most agent communication languages and protocols are based on the Speech Act Theory [7,107], and speech acts are used to distinguish the various types of messages that are sent by and to agents. For example, KQML [37], the best known agent communication languages, is based on speech acts and it defines performatives that are used to identify the

⁵ We refer to software agents as intelligent agents or just agents following the literature and to avoid monotony in writing and reading.

kinds of the agent interactions that take place and to indicate the protocol for message interpretation.

Agent communication languages support several ways for agents to communicate, including direct communication, middle agent-mediated communication, and broadcast. With direct communication, agents handle their own coordination. This method requires an agent to establish direct communication links with other agents and, therefore, requires the agent to be aware of other agents in the system. There are several ways for an agent to get other agents' information. An agent may maintain an acquaintance list, essentially is a list of all agents that it directly knows about. Multi-agent systems that adopt this approach include the Gaia methodology [139] and Sikora and Shaw's [110,111] framework for multi-agent enterprise modeling. An agent may also acquire information about other agents from broadcast messages or from directory services provided by the system. Cost et al. [18] develop an agent-based infrastructure for enterprise integration in which agent name servers provide information about agents in the system. However, direct communication has limitations and supports large numbers of agents poorly; it also requires two agents involved in a communication to use the same agent communication language.

Middle agent-mediated communication is a way to overcome the limitations of direct communication in support of a large number of agents in an open environment (e.g., the Internet). Different types of middle agents have been proposed. A facilitator is an agent that is kept informed about other agents' needs and abilities in a federated system [44]. In such a system, agents communicate only with facilitators, and facilitators communicate with each other. A matchmaker agent helps map agents with supplemental needs and abilities, and a broker agent receives a request from an agent, finds another agent who can process the request, and receives and forwards the result to the requesting agent [24,25]. The middle agent approach has been applied in many multi-agent applications because it allows systems to operate robustly in situations where agents appear and disappear frequently and communications are intermittent. Such applications include, but are not limited to, information retrieval [93,96], e-business [69], and enterprise architecture [112].

Finally, the broadcast method is also used for agent communication. This method of communication is used when an agent wants to send a message to a group of agents, or when the agent does not know who the recipient will be. Broadcast prevents overloading of the network by avoiding multiple copies of the same message being sent to different agents. Broadcast is often used together with direct communication, in which an agent first broadcasts to find appropriate addressees and then communicates with them directly. Two popular architectures for broadcast communication are the contract-net approach and specification sharing [44].

3.2. Multi-agent systems as coordination models

The study of multi-agent systems has its roots in the field of distributed artificial intelligence (DAI). Traditionally, two areas are distinguished within the DAI field: Distributed Problem Solving (DPS) and Multi-agent Systems (MAS) [11]. The DPS paradigm adopts a top-down approach, which divides a complex problem into a number of smaller sub-problems and assigns the sub-problems to multiple problem solvers. A pure DPS system is constructed in such a way that allows all interaction strategies to be incorporated as an integral part of the system. In such a system, all agents are benevolent and cooperative exactly as they are designed to be; therefore, agents can be counted on to act in a way that increases the overall system performance. In contrast, the MAS paradigm takes a bottom-up approach, in which behaviors of individually motivated agents are the focus of the study. Such agents are usually designed by independent designers. What is important to each independent designer is the benefit they can derive from their individually designed agents. Therefore, such agents are also considered as self-interested agents.

Most DAI research has concentrated on techniques for distribution of both control and data to produce coordinated systems [59]. From this perspective, a multi-agent system is a loosely coupled network of problem solvers that interact to solve problems that are beyond the individual capabilities or knowledge of each problem solver [32]. More recently, however, the term "multi-agent system" has taken on a more general meaning, and is used to refer to any system that is composed of multiple autonomous agents and

Table 3
Common coordination mechanisms in MAS

MAS modeling technique	Brief description
Organizational structuring	Organizational structuring gives general, long-term information about the structure of an agent community as a whole, and thus aids the coordination process by specifying exactly the actions an agent will undertake [61]. An organizational structure is useful for modeling the classic master/slave or client/server coordination models for task and resource allocation [98]. The organizational structure can be founded upon the problem decomposition structure [31]. The Distributed Vehicle Monitoring Testbed [23] is an example of a spatial organization in which a community of agents performs a distributed interpretation to track vehicles moving among them.
Meta-level information exchanging	Meta-level information exchange involves agents sending each other control level information about their current priorities and focus in order to reach common conclusions about the problem solving process [30,43]. An agent's decisions are constrained by the actions of other agents. With meta-level information exchanging, an individual agent is expected to maximize its utility function [49,50,122,130]. Recursive modeling is used as a specialized representation for decision-making. An agent builds its decision model based on its own decisions and the decisions of other agents available to it. However, such decision-theoretic methodology has been criticized because it is possible for decision modeling to go on ad infinitum [89].
Multi-agent planning	Multi-agent planning involves finding a multi-agent plan to avoid inconsistent and conflicting actions among distributed agents. The multi-agent plan precisely arranges a priori the tasks each agent will perform and specifies a sequence of actions for each agent. It is different from meta-level information exchange in that it requires agents to reach mutual agreements before acting. However, multi-agent plans can only realistically exist for short time periods of time because of the dynamic environment in a MAS [61]. Also multi-agent planning often requires substantial computational and communication resources because the plan construction has to take into account all possible choice points all agents can possibly reach. Two types of multi-agent planning models have been researched. Centralized multi-agent planning involves a central coordinator for identifying interactions among agents. This is done by integrating partial plans submitted by individual agents into a multi-agent plan [45,47]. On the contrary, distributed multi-agent planning does not use central coordination. Individual agents communicate directly in order to build and update their individual plans and their models of others' plans until all conflicts are resolved [17]. Distributed multi-agent planning has been used in the Distributed Vehicle Monitoring Testbed (DVMT)—a system for testing coordination strategies [80].
Contract net	Contract net protocol is one of the best known and most widely applied coordination mechanisms for task and resource allocation among agents [22,113]. It is modeled on the contracting mechanism used by businesses to govern the exchange of goods and services [59]. The idea behind the contract net protocol is that if an agent cannot solve a problem locally, it will find an appropriate agent to work on the problem. An agent in this approach can take on two roles: manager and contractor. A manager's job is to break a task into its subtasks that can be performed concurrently. The manager then announces the subtasks and requests bids from agents who can perform any of these tasks. After collecting bids from agents, the manager awards the subtasks to the best bidders, who become contractors, and monitors the overall performance of the task. It is possible for contractors to decompose the subtasks further and sub-contract them to other agents. The contract net mechanism provides natural load-balancing as busy agents need not bid, and it is a reliable mechanism for distributed control and failure recovery [58]. However, it does suffer from some limitations: (1) it neither detects nor resolves conflicts, which is one of the key reasons why coordination is needed; (2) agents are assumed to be benevolent and non-antagonistic, which is unrealistic assumption in many real-world problems; and (3) it is rather communication-intensive, the cost of which may outweigh some of its advantages in real-world applications [98].
Negotiation	Negotiation is a communication process for a group of agents to reach a mutually acceptable agreement on some matter [13]. While negotiating, agents first communicate their initial positions, and then move towards agreement by making concessions or searching for alternatives. To negotiate effectively, agents must reason about beliefs, desires, and intentions of other agents [106,119]. Negotiation techniques are classified into three broad categories: game theory-based negotiation, plan-based negotiation, and human-inspired negotiation. Game theory-based negotiation applies to self-interested agents. In this approach, agents are economically rational, and try to maximize their utility functions. Zlotkin and Rosenschein [144] have also extended this work to cover untruthful agents. Plan-based negotiation is a two-stage process [19]. First an agent creates a plan of action that it believes will accomplish its goals. This agent then coordinates its plan with other agents in order to find and resolve possible conflicts. Human-inspired negotiation mimics human negotiation, in which negotiators rely on their past negotiating experience and domain knowledge for conducting successful negotiations.

(continued on next page)

Table 3 (continued)

MAS modeling technique	Brief description
Market mechanism	Market mechanisms enable the construction of an open multi-agent system with a large number of participants in a dynamic manner. Popular market mechanisms include voting, auction, and computational economics. In the voting (social choice) setting, agents choose from a set of alternatives, and then adopt the alternative receiving the most votes [59]. The voting mechanism is simple, equitable and distributed, but it requires significant amounts of communication and organization. The voting mechanism has been proven to improve reliability of decisions in a distributed system [75]. An auction involves two roles: the auctioneer and the bidder. The auction mechanism has been widely adopted in many MAS applications, such as allocation of computing resources in operating systems [131], allocation of bandwidth in computer networks [86], control of building heating [57], and task allocation [48]. Computational economics is the computational study of economies modeled as evolving systems of autonomous interacting agents [123]. In the computational economics setting, everything of interest to an agent is described by prices. Agents bid for goods at various prices, but all exchanges occur at current market price. In a market, agents do not have to behave rationally, but the market will finally reach a competitive equilibrium, which, in some sense, optimally allocates resources and dictates the activities and consumptions of the agents. Patel et al. [103] apply this approach to telecommunication networks to solve the distributed intelligent network load control problem for a multi-service network. Mullen and Breese [94] conduct experiments in designing and building computational economies for distributed operating system resources.

has the following characteristics [1,64,120]: (1) each agent has incomplete information or capabilities for solving the problem and, thus, has a limited viewpoint; (2) there is no global systems control; (3) data are decentralized; and (4) computation is asynchronous.

The MAS research is, for the most part, concerned with coordinating intelligent behavior of a collection of autonomous intelligent agents. It focuses on how agents can coordinate their knowledge, goals, skills, and plans jointly to take actions or to solve problems [11]. The reasons why agents in a system need to be coordinated are as follows [61,98]:

- (1) *Prevention of anarchy or chaos*—a multi-agent system is a decentralized and distributed system with no individual agents having a global view of the system. Without coordination, conflicts, deadlocks, and lack of resources may easily result.
- (2) *Dependencies between agents' actions*—there are dependencies between agents' actions either because local decisions made by one agent may have an impact on the decision of other agent(s) or because there are limited resources in the system that need to be shared.
- (3) *Global constraints*—usually there are global constraints which individual agents must satisfy (e.g., the overall budget constraint of a project).
- (4) *Distributed expertise, resources or information*—individual agents do not have sufficient

competence, resources, or information to solve the entire problem.

- (5) *Efficiency*—even when individuals can work independently, information and knowledge sharing among agents results in a faster solution to the problem.

In order to ensure that agents can work effectively in achieving overall system goals, various coordination techniques have been developed in the intelligent agent and MAS literature. Common coordination techniques include organizational structuring, meta-level information exchanging, multi-agent planning, contract net, negotiation, and market mechanisms [61,98]. While a detailed discussion about these coordination techniques is beyond the scope of this paper, Table 3 gives a brief overview of these techniques for the benefit of the readers⁶ and to facilitate integration of the IBIS and MAS literatures in Section 4.

In short, coordination enables multi-agent systems to work as a single integral unit. From this point of view, multi-agent systems by nature are systems of coordinated problem solvers. This coordination view of multi-agent systems is shown in Fig. 3. A multi-agent system (shown as an external oval) is a collection of agents (shown as rounded rectangles)

⁶ Readers interested in a more detailed discussion about coordination mechanisms used in MAS are referred to Refs. [61,98].

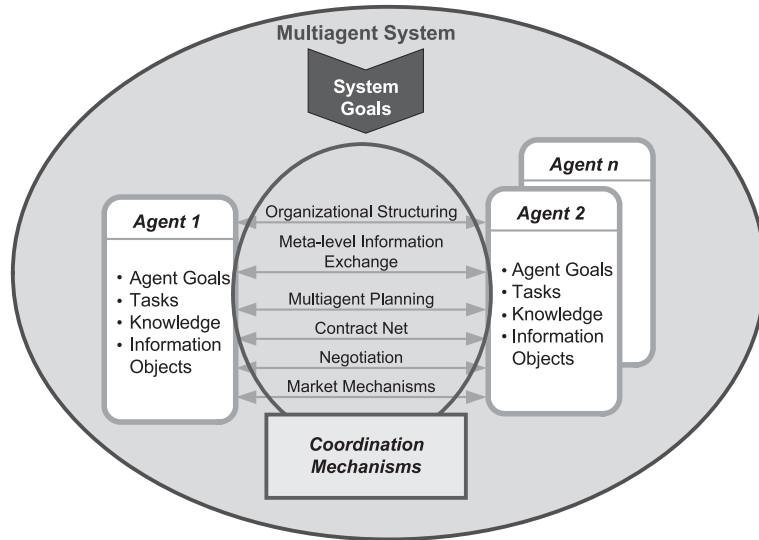


Fig. 3. An overview of multi-agent systems (MAS).

that work together to achieve some overall system goals (shown as an arrow within the system). Sub-goals derived from overall system goals are assigned to individual agents. Each agent possesses certain knowledge about itself and about its environment, and performs some tasks to accomplish its goals. Due to task and resource dependencies and limited knowledge, agents need to be coordinated. The major coordination techniques that are used commonly in multi-agent systems are represented as lines with double arrowheads.

3.3. MAS modeling for enterprise integration

Multi-agent systems are widely applied in different areas in business settings. In this sub-section, we give a brief overview of MAS modeling methodologies, frameworks, and techniques that are particularly suited to the context of enterprise integration. We start with a review of some common and well-known agent architectures and general-purpose MAS development methodologies for domain-independent multi-agent application modeling. Following this review, we focus upon research that provides conceptual MAS models, frameworks, and methodologies for enterprise integration and/or workflow/business process modeling.

The Open Agent Architecture (OAA) [90] is a well-known agent architecture that introduces the

notion of facilitators in multi-agent systems which are essentially entities that are responsible for communication and cooperation between agents in an open environment. RETSINA [121] is a general-purpose modeling framework which proposes goal, role, context and attitude as first class objects for modeling multi-agent systems in an open world. MaSE [26] is another general-purpose MAS development methodology that provides a number of graphically based models for conducting domain-independent MAS analysis and design. During the analysis phase, MaSE includes the notion of goal hierarchy for capturing system goals, and the notion of role models for describing agent behaviors. AUML [10] extends UML for general-purpose modeling of agents and agent interaction protocols by introducing new types of diagrams including agent class diagrams and protocol diagrams. Particular extensions include agent roles, multithreaded lifelines, extended message semantics, parameterized nested protocols, and protocol templates. Prometheus [100] is another general-purpose MAS development methodology that provides support for all activities during the MAS development life cycle including analysis, design, and implementation. For the analysis phase, it includes the constructs of functionality, percept, action, data, and interaction. Functionalities are then grouped into agents. ROADMAP [68] extends the

Gaia methodology [139], a general-purpose agent modeling methodology, to include formal models of knowledge and environment, role hierarchies, and representation of social structures and relationships as well as dynamic changes.

Several authors have proposed a number of MAS modeling frameworks and methodologies for enterprise integration and workflow/business process modeling, and we review the following frameworks and methodologies in this paper: Pan and Tenenbaum's [101] Intelligent Agent (IA)-based framework, Jennings et al.'s [63,65] Advanced Decision Environment for Process Tasks (ADEPT), Sikora and Shaw's [110] enterprise information systems framework, Lin et al.'s [82] enterprise modeling framework, and Yu and Schmid's [141] conceptual framework for agent-oriented and role-based workflow modeling.

Pan and Tenenbaum [101] propose an Intelligent Agent (IA)-based framework to integrate people and computer systems in large, geographically dispersed manufacturing enterprise. In the framework, each IA supports a clearly discernible task or job function, and interacts with each other via a message bus, or through a shared, distributed knowledge base. Humans communicate with this agent system through interface agents called personal assistants. The framework is built by dividing complex enterprise operations into a collection of elementary tasks, and each task is modeled in cognitive terms and entrusted to an IA for execution. Their preliminary experimental results indicate that agent-based systems are a practical way to integrate a complex enterprise.

Jennings et al. [63,65] regard the business process as a community of negotiating agents, and propose the Advanced Decision Environment for Process Tasks (ADEPT) for conceptualizing, designing, and implementing business process management systems. In the framework, various functions of the business process are delegated to a number of autonomous problem-solving agents, and these agents interact and negotiate with each other in order to coordinate their actions and to buy in the services they require. As a conceptual framework for structuring the development of business process management systems, ADEPT identifies and uses key constructs including autonomous agents, negotiation, service provision, service level agreements, resource management, and information sharing.

Sikora and Shaw [110] regard enterprise information systems as consisting of multiple agents with different functionalities, and develop representational formalisms, coordination mechanisms, and control schemes necessary for integrating heterogeneous agents while meeting such performance criteria as overall effectiveness, efficiency, responsiveness, and robustness. Based on this rigorous formal analysis, the same authors propose an agent-based framework for enterprise integration in Ref. [111]. In the framework, each process is modeled as an autonomous agent, which has its own well-defined goals and objectives. Message passing among these agents forms the control structure. Each agent has an internal behavior model, a functional component consisting of procedures/heuristics/strategies, and a protocol for interacting with other agents. The protocols specify what actions an agent will take based on its local state and the messages received. The concurrent execution of the protocols by all agents determines the coordination mechanism used by the agents and the resulting emergence of their behavior.

Lin et al. [82] develop a framework for enterprise modeling using the process hierarchy approach. The authors define an enterprise to be a collection of business entities working toward delivering a product or service to the customer. Each entity performs its independent processes. Processes are organized in a hierarchy. A high-level process consists of a set of low-level processes. In this way, enterprise integration can be modeled at enterprise level, business unit level, and operation level. The authors propose a multi-agent information system (MAIS) for the supply chain network for capturing both the structure and the processes of an enterprise. The MAIS is implemented on the Swarm simulation platform and models the order fulfillment process as one of the core tasks of supply chain networks.

Yu and Schmid [141] propose a conceptual framework for agent-oriented and role-based workflow modeling. A workflow is viewed as a collection of agents interacting with others when they have interdependencies. The authors model a workflow as a set of roles, which are further defined in terms of goals, qualifications, obligations, permissions, and protocols. Interactions among roles are governed by protocols. Roles are then assigned to agents based on the evaluation of qualification and capabilities. Once a

role is assigned to agent, the agent inherits the obligations and permissions specified in that role. Coordination of workflow is achieved by communication among agents.

As can be seen from the above discussion, the MAS paradigm has developed and implemented a variety of coordination mechanisms to ensure that agents within a multi-agent system can work effectively in achieving overall system goals. Further, researchers in the MAS community have also developed conceptual frameworks for multi-agent-based workflow/business processing modeling and

enterprise integration apart from the general-purpose MAS methodologies for domain-independent multi-agent systems modeling. The key constructs for MAS-based enterprise modeling techniques discussed above are summarized in Table 4. Thus, the MAS paradigm seems to be eminently suitable for the development of IBIS systems as coordination is the fundamental integration requirement that integrative business information systems have to support. We next turn our attention to discussing the suitability of the MAS paradigm for integrative business information systems and to synthesizing the two bodies of literature to identify the minimal ontological constructs necessary for analysis and modeling of the multi-agent-based integrative business information systems.

Table 4
Multi-agent systems development methodologies and their key models

MAS enterprise modeling	Key constructs
Pan and Tenenbaum's Intelligent Agent-based framework	Agents (Intelligent Agents and Personal Agents), Activity, Communication
ADEPT	Agents, Agency (Organizational Structure), Services (Tasks and Complex Services), Negotiation.
Silora and Shaw's Model for Multi-agent-based Enterprise Integration	Agents, Performance Measure, Coordination, Control structure (Organizational Structure)
Lin et al.'s Process Hierarchy Model for Multi-agent Enterprise Modeling	Business Unit, Physical Facility, Input/Output, Relation/Interaction, Information Flow/Communication Path, Decision/Strategy, Knowledge Sources, Performance Measures
Yu and Schmid's Workflow Modeling Approach	Workflow, Business Processes, Agents, Roles, Conversations, Protocols, Activities, Obligations, Resources, Permissions, Capabilities, Qualifications
Open Agent Architecture	Agent, Service, Goal, Capability, Facilitation
RETSINA	Agent, Goal, Role, Context, Attitude
MaSE	Agent, Goal, Task, Conversations
AUML	Agent, Interaction, Role
Prometheus	Agent, Functionality, Percept, Action, Data, Interaction
ROADMAP	Agent, Role, Knowledge, Environment, Interaction, Service

4. The MIBIS conceptual framework and foundation ontological constructs

4.1. Suitability of the MAS paradigm for integrative business information systems

An integrative business information system (IBIS) is an information system that integrates multiple work systems of a business enterprise. A multi-agent system (MAS) is a system of software agents that work together to solve problems. From the perspective of architecture, both systems are similar in that they are distributed, decentralized, and composed of system components that need to interact with each other to resolve inconsistencies and dependencies to accomplish goals. Agent-based technology supports complex information systems development by providing natural decomposition, suitable abstraction, and flexibility of management for organizational structure changes [62]. In general, benefits of an agent-based information system include simplification of complex distributed computing, time savings, more and better information, better decisions, improved business processes, and support for accomplishing strategic business objectives [88]. As far as IBIS development is concerned, software agents can increase flexibility of IBIS systems by allowing them to adapt quickly to environmental changes. In addition, by adopting agent technology, IBISs will also be able to integrate heterogeneous technologies and systems as long as agents within the IBIS system can access shared

enterprise ontologies (of organizational definitions and knowledge). Therefore, from the perspectives of architecture and systems development, the multi-agent systems paradigm appears to provide an excellent platform for developing integrative business information systems that are flexible, can integrate multiple technological platforms, and can quickly and easily grow and adapt to changing business environments and needs.

From a modeling perspective, the MAS paradigm needs to support the key constructs that are commonly used in current IBIS analysis and modeling techniques (reviewed above; refer to Section 2.2 and Table 2) and, therefore, appear to be necessary for modeling IBIS applications (these include goals, actors, interdependencies, roles, activities, interactions, workflow, resources, and information). We assess the suitability of the MAS paradigm for modeling and developing IBIS applications by evaluating support of the MAS paradigm for these essential constructs below.

The above review of the two literatures shows that there are conceptual similarities between integrated business enterprises and multi-agent systems in terms of their *goal orientation* (i.e., problem-solving orientation). In an integrated enterprise, actors interact with each other in a number of work systems in a highly coordinated manner to achieve their own and shared goals, thereby contributing to the work system and thereby overall enterprise goals. A multi-agent system is similar to an integrated enterprise because it is essentially a community of autonomous and problem solving agents who interact in computer network systems in a coordinated manner to achieve their own and shared goals, thereby contributing to the achievement of system level goals. In fact, one of the distinguishing features of the MAS paradigm vis-à-vis other modeling approaches, most notably the object-oriented approach, is the goal-oriented behavior of agents within a multi-agent system, which the objects in the OO paradigm lack. Therefore, in terms of the goal construct, the MAS paradigm seems eminently suitable for modeling and developing IBIS applications.

Both integrated business enterprises and MASs are social systems because they are composed of *actors/agents* who are able to act autonomously and interact with other actors/agents and human beings as necessary [64,137]. Furthermore, both human actors and

computational agents are intelligent and adaptive problem-solving entities that can deal with new situations in real time and make decisions based on their prior knowledge and experience. Both actors and agents are also able to take not only reactive but proactive actions as well. Similar to human actors in human social systems and business organizations, intelligent agents in multi-agent systems exhibit both cooperative and self-utility maximizing behaviors towards achievement of their own and system-level goals. From this perspective, the MAS paradigm seems to be a highly suitable approach for modeling integrative business information systems because unlike any other paradigm, agents in the MAS paradigm come closest to mimicking the characteristics of human actors in business organizations: those of intelligence, adaptive problem solving behavior, sociability, and autonomy.

Further, actors within multiple work systems in an integrated enterprise need to be coordinated to manage and resolve *interdependencies* between tasks they perform and resources they use during the course of performance of their tasks. As discussed earlier, multi-agent architectures include well-developed coordination mechanisms to support multifaceted coordination requirements in integrative business systems. For example, organizational structuring provides a central control as a coordination mechanism in MAS, and IBIS applications that wish to use such a centralized control approach for coordinating activities of actors within multiple work systems can use this mechanism. On the other hand, contract net and negotiation approaches in MAS support more decentralized market-oriented coordination mechanisms and use conversations with well-defined performatives from speech acts. Designers and users who wish to implement more market-oriented coordination mechanisms for specific IBIS applications in certain contexts can use these decentralized market-type MAS coordination mechanisms. Further, as discussed above, multi-agent architectures are essentially coordination models (refer to Section 3.2) and, therefore, from the interdependency and coordination perspective also the MAS paradigm seems to be uniquely qualified to model and develop IBIS systems for a wide variety of applications.

Interaction (or communication) among actors, and therefore among work systems, is a fundamental

concept for integrating work systems as it is one of the basic means for coordinating activities of actors and for resolving resource dependencies in an integrated business enterprise. Interaction is also a fundamental concept in the MAS paradigm as agents interact/communicate with one another to share resources, check for task/goal accomplishment, check for availability of other agents for performing certain tasks, negotiate prices and timelines, subcontract tasks to other agents, decide upon future courses of action, etc. *Activities* (or tasks) are also fundamental to both IBIS and MAS because both actors in human organizations and agents in multi-agent communities perform tasks to accomplish their own and organizational/system goals. The notion of *workflow* is also central to both the IBIS and MAS paradigms because a coordinated flow of activities (work) performed by various actors/agents is necessary for achieving organizational/system goals. *Information* is obviously a key construct in both IBIS and MAS domains. A variety of information is required for actors in business enterprises to perform their assigned tasks in order accomplish their goals. Information exchanges also take place between actors within and across work systems in integrated business enterprises as part of their interactions to coordinate their activities and resources. The same is the case with multi-agent systems. Agents within a MAS also use information to perform tasks and exchange information with other agents as part of their interactions. Therefore, from the perspective of interactions, activities, workflow, and information, the MAS paradigm is quite appropriate for the modeling and development of IBIS applications.

The concept of *resources* is indeed a very important construct in integrative business information systems. Resources within business organizations are always finite, and one of the key goals of any business organization is to optimize on resource requirement and utilization to contribute directly to the profitability goal of the organization. When resources are distributed across multiple work systems, improving resource utilization requires their effective coordination to resolve resource dependencies and to share resources across multiple work systems. Thus, resources of a wide variety (e.g., technical skills, documentation, plant and equipment, computer systems, utilities, consumable supplies,

tools, etc.) need to be explicitly modeled in integrative business information systems. Resources are also a key construct in the MAS paradigm as agents contend for and consume finite computational resources such as memory, processor time, communication bandwidth, etc. While the types of resources that agents in the MAS paradigm normally deal with are comparatively limited as compared to business enterprises, any type of physical or intellectual resource can be easily modeled using the MAS paradigm. Thus, the MAS paradigm appears to be suitable for modeling of IBIS systems from the perspective of resources as well.

Finally, *role* has been used as a construct in some IBIS modeling techniques, specifically the Dependency Network Diagrams technique, the Role Activity Diagrams technique, and the WIBO model, to abstract the tasks/activities that need to be performed by real human actors to achieve certain goals. These abstract roles are then assigned to real human actors who are expected to perform tasks/activities included in these roles definition to achieve the goals the roles are designed to accomplish. While the MAS paradigm does not specifically deal with the notion of roles, it can be readily implemented by assigning abstract roles to intelligent agents within the MAS which, as we have discussed above, are similar in a number of ways to human agents. Thus, from the perspective of roles as well, the MAS paradigm provides a suitable approach for modeling and implementation of IBIS applications.

The above discussion clearly indicates that the MAS systems paradigm is eminently suited for modeling and implementing a wide variety of integrative business information systems. Further, the abundance of research on intelligent software agents and multi-agent systems over the last two decades, and the recent focus of this stream of research on business information systems [60,63,65, 66,70,72,102,118,124,142,143] and enterprise integration [82,110,111], make the MAS paradigm undoubtedly a very appropriate platform for integrative business information systems. Therefore, in the next sub-sections we develop a conceptual framework for Multi-agent-based Integrative Business Information Systems (MIBIS) and identify the foundation ontological constructs [15,73,114] for the MIBIS bounded discourse universe.

4.2. The MIBIS conceptual framework

The similarities between the agent construct in the MAS paradigm and the human actors in business organizations in terms of their characteristics and the coordination issues they both face lead us to a conceptualization in which intelligent agents in MAS are used to represent actors in human organizations. The MIBIS framework shown in Fig. 4 reflects this conceptualization. A number of agents (shown as small circles in the figure) representing human actors are contained within the MIBIS system (shown as an oval within the MIBIS environment in the figure). Clusters of agents work closely together to solve larger problems (i.e., accomplish goals). An agent cluster within a MIBIS system (shown as a rounded rectangle in the figure) coincides partially with its related work system in the business enterprise which has been reorganized/restructured as part of the MIBIS analysis and design process to improve organizational effectiveness and system performance.

The overlap between the agent cluster and its related work system depends upon the amount of work (tasks and interactions) that have been delegated from human actors in the business enterprise to intelligent agents within the MIBIS system.

The MIBIS system operates in an environment (shown as the external rectangle in Fig. 4) which is similar to the notion of the environment of an organization and that of a multi-agent system. The MIBIS entity has overall system goals (shown within the arrow in the figure) which describe the problem-solving functions of the system from an environmental user perspective. Users of the MIBIS system define system goals, or the problems that the system will solve (or help solve), during the systems analysis stage. System goals are further decomposed into sub-goals that are assigned to individual agents within the MIBIS system who are responsible for achieving their individual sub-goals. Information resources (the database symbols in the figure) are distributed throughout the MIBIS system either within individual agent

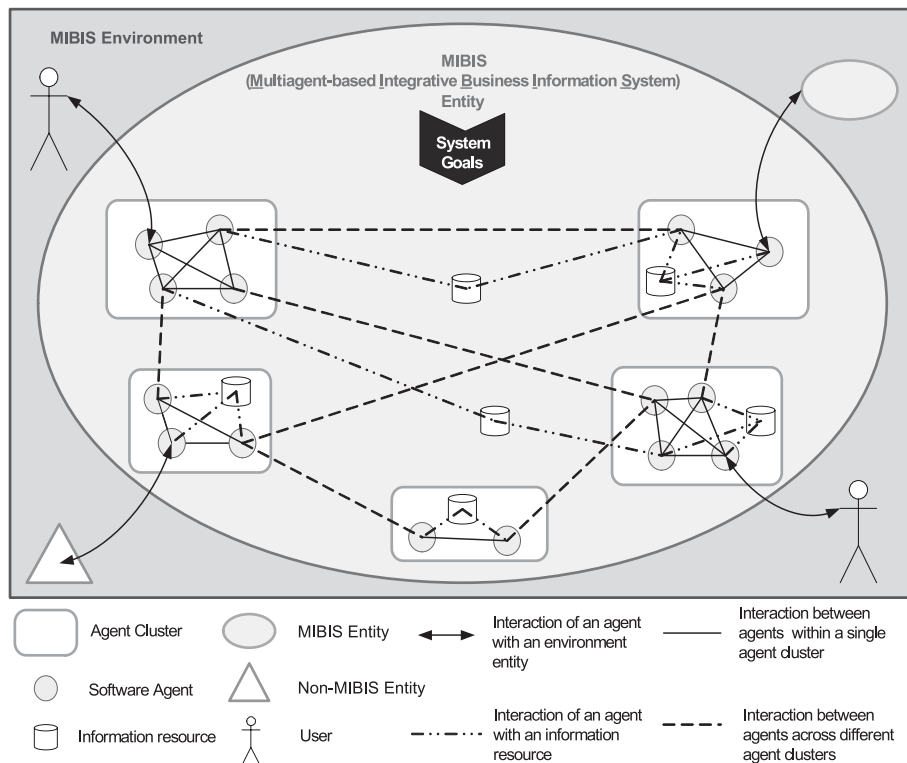


Fig. 4. An overview of multi-agent-based integrative business information systems (MIBIS).

clusters to be shared by agents within those clusters or available as a shared information resource within the MIBIS system which agents from multiple agent clusters can access. Based on the intelligent agent paradigm, agents within a MIBIS system are endowed with private knowledge (not shown in the figure), in terms of factual (declarative) and task execution (procedural) knowledge, that provides them with the capability to solve the problem (goal) assigned to them.

Agents engage in a variety of interactions with other agents in their own cluster, with other agents in other clusters, and with information resources both within and outside their cluster to coordinate their work activities and to accomplish their goals. There are frequent interactions between agents in an agent cluster (therefore, shown as solid lines in Fig. 4) because agents within a cluster usually work as a team requiring more frequent interactions among them to accomplish work system goals as compared to their interactions with agents in other clusters or with external entities. In fact, agents are organized into clusters based on their affinity to other agents in terms of their frequency of interaction with other agents, with a cluster being a collection of high-affinity agents. Interactions among agents also occur across agent clusters (shown as broken lines in the figure) and these interactions help coordinate work across multiple work systems. Interactions between agents and the external environment (shown as solid double-headed arrows in the figure) represent inputs from and outputs to the environment, which consists of environmental entities such as users, other MIBIS systems, and non-MIBIS systems.

Integrative business applications can be implemented as MIBIS applications in a number of domains. For example, the domain of sales order processing in a make-to-stock context is well suited for integration through a MIBIS application. The process involves multiple departments and actors working together to satisfy customers' need. A forecasting clerk in the production planning department may be responsible for forecasting product demand based on past sales information and periodic market research information to ensure that appropriate inventory of products is maintained for sale. When a customer places an order for a

product, the order is reviewed by an order processing clerk in the logistics department. If the product is in inventory, it is shipped; otherwise, the order processing clerk forwards information about the unshipped orders to the production planner in the production planning department who may schedule production of appropriate numbers of units of different products, taking into account the labor and plant capacity, other resources, and time constraints. These tasks performed by human actors can be assigned to intelligent agents in a multi-agent system who will assist the human actors involved in performing them. Thus, a sales order processing system can be implemented as a MIBIS application in which intelligent agents perform tasks, collaborate with other agents to solve problems, accomplish their goals, and assist human actors involved in the process in accomplishing their goals.

A MIBIS application may also be quite appropriate for the procurement process which requires a business organization to coordinate with its suppliers. When the need to purchase is identified, a procurement clerk prepares information such as the materials required, the delivery schedules, the quantities required, and a possible list of suppliers. If new suppliers are needed, the procurement clerk may ask a market research agent to begin the process of identifying and qualifying prospective suppliers. The market research agent gathers information about prospective suppliers and vets them. Once a list of prospective supplier is verified and provided to the procurement clerk, he/she issues a request for quotations for the required materials from the identified suppliers. The bids received from various supplies may have to be evaluated by a bidding clerk. The procurement manager may negotiate the prices and terms with different suppliers and then place an order on the chosen supplier. A MIBIS application for this scenario can be developed for integrating the activities of this process by assigning intelligent agents to various human actors who assist their human counterparts by performing some tasks and interacting with other intelligent agents to coordinate various activities and resources.

We now discuss the ontological constructs required for modeling systems in the MIBIS discourse universe.

4.3. Foundation ontological constructs for the MIBIS discourse universe

Ontological constructs provide the basic language for representing and communicating knowledge in a particular universe of discourse (similar to the notion of an application domain) [15,73,114]. Our interest in this section is to identify a minimal set of orthogonal ontological constructs for the MIBIS discourse universe as these constructs will be essential to understanding and modeling MIBIS applications. Based on our review, analysis, and synthesis above of the integrative business information systems and multi-agent systems literatures, we have identified a set of eight constructs that constitute a minimal set of orthogonal constructs for the MIBIS universe. These eight constructs are agent, role, goal, interaction, task, resource, information, and knowledge, and they are discussed next. Seven of these eight constructs are constructs common to IBIS modeling techniques (reviewed above and summarized in Table 2). Knowledge is included as a construct for the MIBIS universe because it is a core concept in the MAS paradigm. Dependencies and workflow, two constructs common to IBIS modeling techniques, are not included as foundation constructs for the MIBIS universe because they are essentially relationships and can be derived from the eight foundation MIBIS constructs. Table 5 provides information about these constructs in terms of the current IBIS modeling techniques that explicitly provide these constructs, support for them in the MAS literature, and summary discussion about the reasons for including/excluding them for the MIBIS universe. Further discussion about these eight ontological constructs in the MIBIS framework is given below.

As discussed earlier, agents are the problem-solving entity and the central construct in the MAS paradigm. Similarly, actors are the ultimate entity in business organizations that achieve organizational goals. Therefore, the *agent* construct is included as a central construct for MIBIS systems. However, intelligent computational agents in large computational networks such as the Internet are not only transient—they may appear and disappear (die)—but they are also physical incarnations, and there may be several agents written in different programming

languages that solve exactly the same problem. The same is the case with human actors in business organizations. While they are perhaps a little more permanent than intelligent actors, human actors are also transient as they may also leave organizations for a number of reasons (e.g., retirement, resignation, death, etc.). The notion of role, on the other hand, is a more enduring concept in organizations. Organizations define roles, such as that of a purchase manager, a master technician, a woodworker, a financial analyst, etc., as templates for the work these abstract entities will perform and goals they will achieve. Abstract roles are then assigned to physical human actors who play those roles and perform the tasks to accomplish the goals encapsulated within those roles. Also, as discussed above, recently, some techniques have started using the notion of roles as a central concept for the analysis and modeling of IBIS applications. We, therefore, include the notion of *role* as an ontological construct for the MIBIS universe. The two constructs—agent and role—are ontologically orthogonal because one is physical while the other is abstract, the two top level categories in Sowa's ontology [114], and both are included as foundation constructs for the MIBIS universe.

Goal is the third foundation construct for the MIBIS universe. As discussed earlier, both human organizations and multi-agent systems are goal-centric. In fact, goals provide the *raison d'être* for both business organizations and multi-agent systems. Organizational and system goals are divided into sub-goals and are assigned to individual human actors and intelligent agents who accomplish those sub-goals through which organizational and system goals are accomplished. Goal is therefore, included as a foundation ontological construct for the MIBIS universe. As discussed earlier, both IBIS and MAS require coordination to resolve task and resource dependencies, and interactions provide a way for both human actors within organizations and agents within a MIBIS system to coordinate tasks and resources and to share information with one another. Therefore, *interaction* is also included as a foundation construct for the MIBIS universe.

The next fundamental construct for the MIBIS universe is the notion of tasks. Both human actors in business organizations and intelligent agents with a MIBIS system perform tasks to accomplish their

Table 5

Foundation ontological constructs for the MIBIS bounded discourse universe

Construct	Construct explicitly provided in the following IBIS modeling approaches	Support for the construct in the MAS paradigm	Reasons for including/excluding the construct as a foundation ontological construct for the MIBIS bounded discourse universe
<i>Foundation ontological constructs proposed for analysis and modeling of MIBIS applications</i>			
Agent ^a	○ Actor Dependency Model ○ ECHOES Modeling Language ○ Role Activity Diagrams ○ Workflow Intelligent Business Object ○ Action Workflow System ○ SAMPO Model ○ DEMO Model ○ Weigand's Model ○ TOVE	Agents are the fundamental concept in the MAS paradigm. There are multiple agents in a multi-agent system. However, no agent possesses the knowledge or the capabilities to understand and solve the entire problem.	Human actors in business organizations, to be modeled as agents in MIBIS applications, are one of the key entities (others are machines and computational systems) that perform tasks to accomplish goals, and need to be modeled in MIBIS applications as intelligent agents. Therefore, this construct has been included as a foundation construct for MIBIS modeling.
Role	○ Dependency Network Diagrams ○ Role Activity Diagrams ○ Workflow Intelligent Business Object ○ TOVE	While the multi-agent systems paradigm does not provide explicitly for the notion of a role, each agent in a multi-agent system plays a particular to solve partial problems.	Every actor within an organizational work system plays one or more roles, which are abstractions for the tasks that need to be performed to accomplish certain goals. The notion of roles is more fundamental than human actors in business organizations, and the same role may be assigned to another actor if the first actor leaves or becomes unavailable. Therefore, this construct has been included as a foundation construct for MIBIS modeling.
Goal	○ Actor Dependency Model ○ Dependency Network Diagrams ○ Weigand's Model ○ TOVE	A multi-agent system has an overall goal. Individual agents are designed to achieve one or more goals, but all agents working in concert in a MAS are expected to achieve the overall system goal.	Goal is a fundamental concept in both business organizations and MAS. Goals define the raison d'être for both business organizations and multi-agent systems. Larger organizational goals are sub-divided into smaller goals and actors within work systems have goals that they need to accomplish. The same is the case with MAS. Therefore, goal is included as a foundation concept for MIBIS modeling.
Interaction ^b	○ Role Activity Diagrams ○ Workflow Intelligent Business Object ○ DEMO Model ○ Weigand's Model ○ Lind and Goldkuhl's Model ○ TOVE	Interaction and conversations are the key means for sharing information between agents to coordinate their activities and goals. Agent interaction languages and protocols are quite well studied and developed in both the cooperative and competitive MAS environments.	Interactions are fundamental to business organizations in general, but are more crucial in the context of integrative business information systems. Interactions occur both within and across work systems in business organizations to solve inconsistencies or conflicts caused by goal dependencies, task dependencies, and resource dependencies. Interactions are also a fundamental concept in the MAS paradigm. Therefore, interaction has been included as a foundation construct for the MIBIS universe.
Task ^c	○ Actor Dependency Model ○ Dependency Network Diagrams ○ Role Activity Diagrams ○ Workflow Intelligent Business Object ○ XRL ○ Action Workflow System ○ SAMPO Model	Task is another fundamental concept in the MAS paradigm. Each agent is responsible for performing tasks to solve partial problems. MAS supports task synchronization, task allocation, task sharing, and result sharing.	Tasks are fundamental to business organizations and practically all IBIS modeling techniques provide for modeling of tasks, activities, or processes. Tasks are also central to the MAS paradigm. Therefore, this construct has been included as part of the foundation constructs for the MIBIS universe.

(continued on next page)

Table 5 (continued)

Construct	Construct explicitly provided in the following IBIS modeling approaches	Support for the construct in the MAS paradigm	Reasons for including/excluding the construct as a foundation ontological construct for the MIBIS bounded discourse universe
Resource	○ DEMO Model ○ Weigand's Model ○ Lind and Goldkuhl's Model ○ SEAM ○ TOVE ○ Basu and Blanning's Model ○ Actor Dependency Model ○ Workflow Intelligent Business Object ○ TOVE ○ Basu and Blanning's Model	While resources are not an explicit construct in the MAS paradigm, agents do utilize computational resources. Agent coordination mechanisms in multi-agent systems also acknowledge and incorporate resource constraints in their algorithms.	Resource is one of the most important concepts in business organizations. Resources are finite and resource dependencies occur that need to be resolved. These dependencies are more important to be coordinated to effectively integrate work systems. Therefore, resource is included as a foundation construct for MIBIS modeling.
Information ^d	○ ECHOES Modeling Language ○ DEMO Model ○ SEAM ○ Basu and Blanning's Model	Agents receive and provide information from and to a variety of sources including other agents and external information sources.	Information is also a fundamental concept in business organizations and is now regarded as vital corporate asset. It is glue that binds modern-day decentralized and distributed organizations and is, therefore, included as a foundation construct for MIBIS modeling.
Knowledge	○ Actor Dependency Model (modeled as know-how)	Knowledge is an integral part of the intelligent agent and multi-agent systems paradigm. Agents possess knowledge and act based on their own knowledge, beliefs, desires, and intentions.	While knowledge is not a common construct in current IBIS modeling techniques, we have included it as a foundation ontological construct for the MIBIS discourse universe because the MAS paradigm is an integral part of this universe and knowledge is a basic construct in that paradigm. Further, knowledge is also fundamental to human actors in business organizations, as they also possess knowledge in terms of know-how, skills, expertise, and prior experiences that they bring to bear on their tasks and problem solving activities.
<i>Constructs Used in IBIS Modeling but not included as Foundation Constructs for MIBIS Analysis and Modeling</i>			
Dependency ^e	○ Actor Dependency Model ○ Dependency Network Diagrams	The notion of dependency is central to the MAS paradigm. Agents in a multi-agent system solve only partial problems, have a limited view of the overall goal, and are dependent on other agents to solve larger problems. As discussed, MAS are essentially coordination models to resolve dependencies among agents.	While central to integrative business information systems, dependencies can be modeled as a relationship between tasks, resources, and agents, which are the foundation constructs identified for the MIBIS universe. Therefore, dependency is not included as an explicit foundation construct for MIBIS modeling.
Workflow	○ Workflow Intelligent Business Object ○ XRL ○ Action Workflow System ○ Weigand's Model ○ SEAM ○ Basu and Blanning's Model	This is not a central construct in the MAS paradigm. However, agents perform task in certain sequences and experience interdependencies among tasks.	The notion of workflow is central to integrative business information systems. However, workflow, or task execution patterns of a wide variety, can be modeled as relationships among tasks performed by one or more agent. Therefore, this construct is not included as an explicit foundation construct for MIBIS modeling.

goals. While sometimes there is some confusion in distinguishing between goals and tasks (e.g., should we classify “order inventory” be a goal or a task?), the two concepts are ontologically distinct and orthogonal. At a given level of analysis and abstraction, goals always define “what” needs to be accomplished while tasks always define “how” the what (goals) will be accomplished. Further, goals are always the output of the entity under consideration defined in conjunction with the environment of the entity, while tasks are the input and are determined solely by the entity, obviously in a manner that results in efficiency. The environment is not concerned with the tasks of the entity as long as it performs some tasks to accomplish the agreed upon goals. Similarly, tasks are also considered ontologically orthogonal to the notion of interactions in our analysis for the MIBIS universe. A task is a unit of work that is performed solely by an individual agent working by itself, while interaction is a unit of work that can only be performed by two agents working in concert to coordinate their dependencies or to share information. Therefore, *task* is included as a foundation construct for the MIBIS universe.

Resource is also included as a foundation construct for the MIBIS universe because effective coordination cannot occur without resolving task–resource and resource–resource dependencies that are common to integrative business information systems. Resource is orthogonal to all the constructs discussed so far. *Information* and *knowledge* are the remaining two constructs included for the MIBIS universe. While both information and knowledge need no justification for inclusion as modeling constructs for the MIBIS universe—they have been recognized as the key abstract resources in the knowledge economy and are crucial to the analysis and operation of MIBIS systems—the distinction between these two constructs needs to be highlighted to ensure that all the

constructs in the MIBIS universe are indeed orthogonal. Our intention here is not to settle the debate about the ontological distinctions between these two constructs—it will continue in the philosophy and other related disciplines, as it has for centuries. Rather, we define knowledge as “justified true belief” drawing from traditional epistemology but, following a recent and prominent viewpoint about organizational knowledge [97], we emphasize knowledge to be a personal justified belief of an entity (a human or computational agent) rather than an absolute and static true belief (for all entities at all times). As beliefs are internal to entities, because it is they who hold those beliefs, knowledge is treated as internal to an entity. However, information comes from a variety of external sources and is absorbed by human beings using their five senses and by machine agents using their input devices. Using new information from their environments and justification and truth evaluation processes, human and machine agents may revise their current beliefs and form new beliefs, thereby adding to, restructuring, or changing their knowledge [29,97]. Thus, information in our conception is purely external. Because knowledge is central to multi-agent systems—agents are intelligent, i.e., they possess knowledge—we include both information and knowledge as the foundation constructs for the MIBIS universe.

5. Discussion

As discussed in the previous section, the MIBIS conceptual framework identifies eight ontological constructs and they can be used to model the goals, actors, tasks, knowledge, information, interdependencies, and interactions of some of the work systems of a business organization that need to be integrated using

Notes to Table 5:

^a The agent construct refers to both the notion of a human actor in business organizations and an intelligent agent in multi-agent systems.

^b Current IBIS modeling techniques use the notions of conversation and interactions which are conceptually similar. The interaction construct, therefore, includes both the interaction and conversation constructs used in current modeling techniques.

^c Current IBIS modeling techniques use the notions of activity, task, process, business process, act, or action which are conceptually similar. The task construct, therefore, includes the task, activity, process, business process, act, or action constructs used in current modeling techniques.

^d The information construct includes both the information store (also called information bank in some techniques) and information flow constructs used in current modeling techniques.

^e The terms dependency and interdependency are used synonymously in this paper.

a multi-agent-based integrative business information system. Thus, these constructs provide support for enterprise integration by allowing a coherent modeling of these fundamental concepts of an integrated business enterprise.

As discussed earlier, an integrated business enterprise is essentially a social system in the sense that it is composed of actors who are able to act autonomously and interact with other actors as necessary [64,137]. The MIBIS framework provides the notion of agent as a first-class entity using which the actors of an integrated enterprise can be modeled. Further, the concept of role in the MIBIS framework enables modeling of stable organizational structures of an integrated enterprise. Abstract roles are assigned to agents who are expected to perform tasks included in these roles definition to achieve the goals the roles are designed to accomplish. The tasks the roles are expected to perform can be modeled using the task construct provided in the MIBIS framework. Similarly, the resources an agent will use to perform a task can be modeled using the resource construct while the information that a role will access and/or create during the course of performance of a task to accomplish a goal can be modeled using the information construct provided by the MIBIS framework.

From the perspective of dependency modeling, the MIBIS framework supports goal dependencies, task dependencies, and resource dependencies. Goal dependencies can be captured by the goal construct in terms of a goal tree. An overall enterprise goal can be decomposed recursively into sub-level goals. Leaf goals of the goal tree can then be assigned to individual roles as their responsibilities. Roles in turn can be instantiated by agents. The MIBIS framework, thus, ensures that agents not only work in a coordinated manner and contribute to the overall system goal, it also does not infringe upon the autonomous and proactive nature of agents and agents can attempt to maximize their self interests under the constraints of the goal tree.

Task dependencies and resource dependencies in an integrated enterprise can be represented using the knowledge construct provided by the MIBIS framework. Task dependencies may exist within or between roles. Within-role task dependencies occur

as one task that a role is supposed to accomplish may depend upon one or more tasks that the same role is assigned to complete, and these can be represented as task structure using the knowledge construct. Between-role task dependencies occur as one task that a role is supposed to accomplish may depend upon one or more tasks that other role(s) are supposed to accomplish. These dependencies can also be represented by way of task execution constraints within the knowledge constructs. These constraints should not identify the specific tasks that are to be accomplished by other roles as that will violate the basic principle of autonomy of agents. Rather, completion of the prior required tasks by other roles should be signaled by message flows between roles to resolve task dependencies and it is the message flow information that should be captured and represented as a constraint using the knowledge construct. Resource dependencies within and among roles and tasks can also be similarly captured and represented using the knowledge construct. The knowledge construct specifies which task performed by a particular role requires what resources. The knowledge construct can also be used to model the contention of resources and availability of resources can be signaled using messages and this message flow information (e.g., what message should be expected and from whom when a required resource becomes available) can be modeled using the knowledge construct.

Finally, for modeling the messages alluded to above that are required for resolving various kinds of dependencies and coordinating the goals and tasks assigned to various roles, the MIBIS framework provides the interaction construct. Interaction enables agents to communicate with one another to share resources, check for task/goal accomplishment, check for availability of other agents for performing certain tasks, negotiate prices and timelines, subcontract tasks to other agents, decide upon future courses of action, etc.

As seen from the discussion above, the MIBIS ontological constructs are semantically quite coherent with the realities of integrated business enterprises and, thus, allow for conceptualization of all the fundamental and crucial elements of such enterprises in a very effective manner.

6. Conclusions

Enterprise integration through integrated business information systems (IBIS) is one of the most important issues facing business enterprises in this age of hyper-competition. A high degree of agility and responsiveness, a competitive necessity, can only be achieved when enterprise work systems and supporting information systems are properly integrated. Our review above indicates that the multi-agent systems (MAS) paradigm provides an excellent architecture, technological platform, and modeling approach for developing and implementing IBIS systems that are flexible and can easily and quickly grow and adapt to changing business environments. Our synthesis of the IBIS modeling and MAS literatures has led us to propose a multi-agent-based integrative business information systems (MIBIS) conceptual framework and a unified set of eight orthogonal ontological constructs that are minimally required for any conceptual modeling grammar [132] for the MIBIS universe. Future research should develop formal semantics and axioms for the eight constructs identified in this paper and relationships among them as part of a formal conceptual modeling grammar for the MIBIS universe.

Acknowledgements

The authors are grateful to Kaushal Chari and participants at the MSS Colloquium at SUNY Buffalo for providing helpful suggestions during the course of development of this paper. The authors are also grateful to the two anonymous reviewers for their invaluable comments and suggestions, which have greatly helped improve the quality of this paper.

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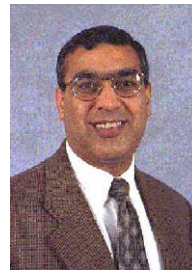
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